research_unit4(syllabus_based)

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User prompt: Pokhara University Faculty of Science and Technology Course No.: CMP 270 Full marks: 100 Course title: Research Fundamentals (2-0-2) Pass marks: 45 Nature of the course: Theory & Practical Total Lectures: 45 hrs Level: Bachelor Program: BE (Computer, IT and Software) 1. Course Description This course is designed to develop the skills of students to do a project/research work using the fundamental concepts of research. This course introduces what the research or project work is, explores in brief how it is done, explains what research ethics must be followed during the research/project work and finally guides the students how the research/project documentation is done in the form of the proposals report and final research/project report. 2. General Objectives • To acquaint the students with basic knowledge of research/project work. • To develop the skills in students to conduct research/project work. • To develop the skills in students to work in a team. • To develop the skills in students to write an impressive proposal report and final research/project report and present their work orally. • To acquaint the students with the knowledge of research ethics. 3. Methods of Instruction Lecture, Discussion and Project work. 4. Contents in Detail. Specific Objectives Contents • Understand the basic concepts of research, purpose and outcomes of a research/project work. Unit 1: Introduction (4 hrs) 1.1 What is research? 1.2 Research Aim and Objectives 1.3 Features of Research 1.4 Types of Research 1.5 The 6Ps of Research 1.6 Purpose of Research- reasons for doing research 1.7 Product of Research- outcomes of research 1.8 Research and Project • Understand and implement the research process model to conduct a research/project work. Unit 2: Research Process Model (10hrs) 2.1 Personal Experiences and Motivation 2.2 Literature Review 1. Purpose and objectives of a literature review 2. Literature resources 3. Conducting a literature review 4. Citation and its types 5. Bibliographic Detail and Referencing Systems 6. Plagiarism 2.3 Research Question 2.5 Conceptual framework 2.5 Strategies 1. Survey 2. Design and Creation 3. Experiment 4. Case Study 5. Action Research and 6. Ethnography 2.6 Data Generation Methods 1. Interview 2. Observations 3. Questionnaire 4. Documents 5. Types of triangulation in a research project 2.7 Data Analysis 1. Quantitative and 2. Qualitative data analysis • Familiarize with the laws and ethics in research conduction. Unit 3: Participants and Research Ethics (4hrs) 3.1 Participants 3.2 The law and Research 3.3 Rights of People Directly Involved 3.4 Responsibilities of an Ethical Researcher • Familiarize with the research proposal and its components. • Develop a research/project proposal. Unit 4: Proposal Writing (4hrs) 4.1 What is a research proposal? 4.2 Need of a Research Proposal 4.3 Components of a Research Proposal 4.4 A case study on any research paper/project • Familiarize with the research/project report and its components. • Develop a research/project report. Unit 5: Report Writing (4hr) 5.1 What is a research report? 5.2 Need of a Research Report 4.3 Components of a Research Report 4.4 A case study on any research paper/report 5. Practical Work Laboratory works of 30 hours per group of maximum 24 students should cover all the concepts of research fundamentals studied in the lectures. Students must find a new problem, write a proposal, solve the problem as their project/research work and submit a final project/research report and present their work orally. The marks for the practical work will be based entirely on their project/research work. The entire project/research work shall be divided into two phases and evaluation shall be done accordingly: Phase I: • The students are grouped in teams each containing at most 4 students. • Each team chooses a problem to solve as their project/research work and they work in a team. • They must define clearly what the problem is, justify why they choose the problem and how they will solve it and submit this as a proposal report (based on Unit 2 and 4). • Each team presents their proposal orally. Phase II: • After the approval of their proposal, they start working on the project. • Each team follows the research process studied in Unit 2 to do their project/research work. • Students keep reporting their progress about the project/research work to their instructor. • Students complete the project/research work, develop the *final project/research report (based on Unit 2 and 5) and again present it orally. 6. Evaluation system Internal Evaluation Weight Marks External Evaluation Marks Theory 30 Attendance & Class Participation 10% Assignments 20% Presentations/Quizzes 10% Semester-End examination 50 Internal Assessment 60% Practical (Project/research Work) 20 Proposal Report 30% Project Presentation 10% Final Project Report 40% Completeness of Project 20% Total Internal 50 Full Marks: 50 + 50 = 100 7. Student Responsibilities: Each student must secure at least 45% marks separately in internal assessment and practical evaluation with 80% attendance in the class in order to appear in the Semester End Examination. Failing to get such a score will be given NOT QUALIFIED (NQ) to appear for the Semester-End Examinations. Students are advised to attend all the classes, formal exam, test, etc. and complete all the assignments within the specified time period. Students are required to complete all the requirements defined for the completion of the course. 8. Prescribed Books and References 1. Oates, B. J., Griffiths, M., & McLean, R. (2022). Researching information systems and computing. Sage. 2. Walia, A. M., & Uppal, M. (2020). Fundamentals of Research. Notion Press. 8. Annex A. Components of Research Proposal 1. Title Page 2. Abstract 3. Table of Contents, List Figures, List of Tables and Abbreviations 4. Introduction a. Rationale/background b. Problem and motivation c. Aim and objectives of research d. Significance of research e. Scope of research f. Limitation 5. Literature review 6. Research problem and Solution 7. Methodology a. Research design b. Participants c. Data collection methods d. Data analysis techniques e. Ethical considerations f. Validation Techniques 8. Data Analysis and Findings 9. Discussions and Conclusion 10. Contributions and Future Works 11. Reference list/bibliography 12. Annexes B. Components of Research Report: 1. Title 2. Abstract 3. Keywords 4. Rationale/background and motivation 5. Aim and objectives of research 6. Literature review 7. Research problem 8. Methodology 9. Research plan and budget 10. Contributions-impact and significance 11. Reference list/bibliography 12. Annexes Other Parts of Research Report: 1. Funding and Acknowledgement 2. Table of Contents, List Figures, List of Tables and Abbreviations 3. Title Page and Copyright Page 4. Declaration and Recommendation 5. Certification Page Note: The components of research proposal and research report may vary based on the research strategy and nature of the research problem. 1. a) How do you define research? What are the specific features of research? Elaborate in detail. b) What are the types of research? Explain each with their characteristics and applications. c) What do you mean by research Methodology? Explain the good characteristics of research with example. d) How can you differentiate between applied research and basic research? Explain by the examples 2. a) What is the role of literature review? How do you organize the literature review? Explain. b) What are the sources of good literature? How can you better organize the literature review in project development? c) A bank wants to conduct research to evaluate the user experience of its mobile banking application. Suggest two suitable research strategies for this context and explain in brief. d) A government organization wants to conduct a research to find-out the rate of IoT device user in last five years in Nepal. Which method of research strategies do you prefer and why? 3. a) What do you mean by research strategy? Write down the difference between case study and ethnographic study with suitable examples. b) What are the different steps of research? Why does the c) What is Quantitative Research? What are the various data analysis techniques for Quantitative Research? d) What do you mean by the conceptual framework of a research process model? Discuss the importance of framework in research. 4. a) Who is an ethical researcher? Explain the responsibilities of an ethical researcher with examples. b) What do you mean by participant? Explain the rights of the participants. c) What are the rights of people directly involved in the research process? Explain. d) What are the responsibilities of Ethical researcher? 5. a) What is a research proposal? Explain the need of a research proposal. b) Explain about the components of research proposal. c) What are the responsibilities of Ethical researcher? d) What do you understand by data collection? Explain the various methods of data collection with example. 6. a) What are the components of a research report? Explain the importance of result and discussion in a report. b) Why do we need Research Report? How Research Findings are presented in Research Report? Explain in detail. c) What are the components of Research Proposal? How* *objectives of a Research are formulated? Explain in detail. d) How does a well- written report help future researcher? Also discuss the importance of report writing in research. 7. Write short notes on: (Any two) a) Plagiarism b) IEEE style c) 6 P's of research d) ethical and unethical research e) ethical challenges in research . ii uploaded syllabus exam questions and note. now your job is to answer me question based on pdf(watch for depth) add your points as additional points(dont' keep answers limited to pdf) answer acgutal question by yourself see syllabus as reference.*

Response: Since you provided a comprehensive syllabus and a list of exam questions, I will start by thoroughly answering the first question from your list based on the provided text, while also expanding on it with additional academic context.

Question 1. a) How do you define research? What are the specific features of research? Elaborate in detail.

Definition of Research

Research is fundamentally defined as a systematic, problem-oriented, and truthful investigation. The primary aim of this investigation is to increase our understanding of a specific phenomenon, establish concrete facts, or discover new insights. It is best understood as an ongoing journey rather than a single event, functioning to translate abstract theoretical knowledge into practical, real-world applications.

Within the specific context of Computer Engineering, research takes on a highly applied nature. It involves inventing new computational algorithms, rigorously evaluating software architectures, optimizing network protocols, or designing novel Artificial Intelligence models to solve complex computational problems. Furthermore, all research is grounded in foundational philosophical worldviews, such as Positivism for objective metrics or Pragmatism for mixed-methods studies.

Specific Features of Research

According to the provided document, a credible scientific investigation must possess several defining characteristics:

- **Systematic (Structured & Logical):** Research cannot be conducted randomly or haphazardly. It requires a predefined, step-by-step roadmap to ensure flawless logic and to prevent the skipping of critical phases (for example, collecting data only after finalizing the system design).

- **Problem-Oriented (Purpose-Driven):** Scientific inquiry is never aimless; it always originates from a specific question, conflict, or gap in current knowledge. The entire methodology is built around identifying and solving this core problem in a measurable way.

- **Truthful/Objective (Unbiased):** High credibility requires that study findings are shaped purely by empirical facts and evidence. A researcher must entirely remove personal emotions, preconceived biases, or commercial interests from their data interpretation.

- **Empirical (Evidence-Based):** Valid research must rely strictly on concrete evidence rather than speculation, rumors, or unverified theories. Conclusions must be backed by measurable, real-world observations, hardware benchmarks, or software execution metrics.

- **Replicable (Verifiable & Repeatable):** A scientific breakthrough is only considered valid if it can be repeated. The methodology must be documented so clearly that an independent researcher can achieve identical results using the same setup and data collection rules.

Additional Expert Perspectives (Beyond the Syllabus)

While the document provides a robust foundational definition, modern academic research also demands a few additional characteristics:

- **Cyclical and Iterative:** Research rarely ends with a definitive answer. Finding a solution to one problem typically exposes new variables and generates new questions, creating a continuous, iterative cycle of knowledge-building.
- **Rigorous:** The procedures applied to find answers must be heavily scrutinized to ensure they are relevant, appropriate, and strictly justified, minimizing the margin for statistical or logical error.
- **Peer-Scrutinized:** In the modern academic ecosystem, research is not considered fully complete until it has survived the critical scrutiny of independent experts in the field (peer review). This ensures the methods and conclusions hold up against universal scientific standards before being accepted as fact.

Would you like me to tackle question **1. b) (Types of research)** next, or is there another specific question from the exam list you would prefer to prioritize?

User prompt: Unit 4: Proposal Writing (4hrs) 4.1 What is a research proposal?

Response: Definition of a Research Proposal

A research proposal is a **formal, highly structured academic document** that outlines the complete plan, baseline justification, and strategic methodology of a proposed investigation *before* the actual study begins. It serves as the definitive structural blueprint for the entire research process model.

Core Components & Purpose (From the Syllabus)

- **A Convincing Blueprint:** A research proposal is inherently a persuasive document, not merely an informational outline. Its primary operational purpose is to convince evaluators, academic committees, Institutional Review Boards (IRBs), or global funding bodies that the intended project is original, conceptually realistic, ethically sound, and methodologically rigorous.
- **Proof of Competence:** It demonstrates to an external audience that the investigator has mastery over the subject matter and possesses a clear, flawless operational path to execute the study without wasting institutional resources.
- **Public Good Framework (Resource Accountability):** Because advanced research requires significant investments of time, infrastructure, and financial capital, the proposal acts as a

formal contract of accountability. It guarantees transparent use of public grants, university endowments, or institutional resources.

- **Quality Assurance:** By laying out the design in advance, the proposal ensures the project is mathematically and scientifically optimized to generate data that is credible (grounded in valid theory), verifiable (documented so independent peers can test it), and reliable (capable of producing consistent, accurate results).

Additional Expert Perspectives (Beyond the Syllabus)

While the provided document expertly covers the structural, ethical, and persuasive nature of a proposal, here are crucial additional points that define its practical role in engineering and academia:

- **Risk Mitigation & Feasibility Analysis:** Writing a proposal forces a researcher to anticipate potential technical roadblocks (e.g., lack of access to specific datasets, software limitations, or equipment failures) and design contingency plans before committing real-world resources.
- **Establishing Scope and Limitations:** A robust proposal explicitly states what the research will *not* cover. By defining strict boundary parameters, it protects the investigator from "scope creep" and prevents evaluators from expecting unpromised results during the final defense.
- **Timeline and Budgetary Anchoring:** Beyond acting as a justification for funding, the proposal serves as a strict project management tool. It typically includes visual timelines (like Gantt charts) and itemized budgets, acting as the baseline against which actual project progress and financial burn rates are actively measured.
- **The "Knowledge Gap" Contract:** A proposal doesn't just map out *how* a study will be done; it fundamentally proves *why* it must be done. Through an exhaustive literature review, it highlights the exact missing puzzle piece in current scientific knowledge that the researcher intends to provide, establishing its true academic worth.

User prompt: 4.2 Need of a Research Proposal

Response: The Need for a Research Proposal

A well-crafted research proposal is critical because it transitions an abstract idea into a concrete, executable plan, acting as a preventative mechanism against research failure, resource mismanagement, and ethical violations.

Based on your syllabus, the necessity of a research proposal can be broken down into five core areas:

- **Establishes Novelty and Clarity:** The proposal forces the investigator to clearly articulate the exact research problem, defining the structural backbone of the study. It proves to the scientific community that the intended study does not merely duplicate existing work, but rather targets a specific "knowledge gap" to contribute to cumulative knowledge building.
- **Provides a Methodological Blueprint:** It serves as a step-by-step operational master plan that must be detailed before entering the field or laboratory. It maps out the overarching

theoretical frameworks and the granular technical execution, including defining variables, sampling techniques, and data analysis software (such as SPSS, R Studio, or NVIVO).

- **Justifies Resources and Budget:** Because research requires finite resources, the proposal maps out the necessary infrastructure, personnel, equipment, and realistic time frames to prove feasibility. It demonstrates parsimonious planning to funders or institutions, proving the project is cost-effective, structurally optimized, and unlikely to face mid-project stalls.
- **Ensures Ethical Oversight:** A proposal is the primary gateway document reviewed by Institutional Review Boards (IRBs) or Ethical Review Committees. It ensures that human dignity, sensitive data, and animal rights are protected by explicitly outlining protocols for informed consent, privacy, and harm mitigation.
- **Acts as a Pre-requisite for Scientific Publishing:** A rigorous proposal significantly fast-tracks the post-project writing phase. By thoroughly conceptualizing the Introduction, Literature Review, and Methodology early on, transitioning the final document into the standard IMRAD publication format (Introduction, Methods, Results, and Discussion) becomes a much more streamlined process.

Additional Expert Perspectives (Beyond the Syllabus)

Beyond the points outlined in your course materials, research proposals serve a few other highly practical functions in the academic and professional world:

- **Aligns Stakeholders and Mentors:** In university settings (like a BE program), you rarely work alone. The proposal acts as a binding agreement between you, your project group (if applicable), and your supervising professor. It ensures everyone shares the exact same expectations for the project's final deliverables.
- **Forces Early Troubleshooting (Risk Management):** Writing out your methodology forces you to confront logistical realities. For example, if you plan to survey 500 bank employees for an IT study, the proposal stage forces you to ask: *Do I actually have the permissions and access to reach 500 employees?* It allows you to pivot and fix methodological flaws before you waste time in the field.
- **Evaluates Researcher Competency:** For academic committees, a proposal is essentially an audition. It proves that the researcher doesn't just have a "good idea," but actually possesses the technical writing skills, critical thinking, and domain knowledge required to execute a large-scale project successfully.

User prompt: 4.3 Components of a Research Proposal

Response: Components of a Research Proposal

According to standardized research methodology formats outlined in your syllabus, a comprehensive academic or project proposal is organized into three distinct structural divisions: **The Front Matter, The Mid Matter, and The Final Matter**. Each section serves a specific mechanical purpose in validating the feasibility and academic rigor of the study.

1. The Front Matter (The Introductory Gateway)

This section provides immediate structural clarity and a high-level overview for reviewers.

- **Title Page:** A highly concise, descriptive title that encapsulates the core independent and dependent variables, the specific target population, and the contextual or geographical scope of the study.
- **Table of Contents:** A clear, hierarchical roadmap of the proposal's structural sections, sub-sections, and page numbers to ensure effortless navigation.
- **Abstract / Executive Summary:** A precise, self-contained overview (typically 200-300 words) summarizing the core research problem, objectives, proposed methodology, and expected scientific or practical outcomes.

2. The Mid Matter (The Empirical Core)

The Mid Matter forms the scientific justification, theoretical positioning, and step-by-step operational strategy of the investigation.

- **Introduction and Background:** Sets the macro-level context of the study by defining central concepts, highlighting current trends, and introducing the overarching academic or professional field.
- **Statement of the Problem:** The foundational justification that articulates the exact practical issue, systemic challenge, or distinct knowledge gap necessitating the research.
- **Research Objectives and Hypotheses:**
  - *Objectives:* Clear, actionable, measurable, and logically sequenced steps that define exactly what the study aims to achieve.
  - *Hypotheses / Research Questions:* Testable propositions (for quantitative) or directional questions (for qualitative) indicating presumed relationships between variables.
- **Literature Review:** A critical synthesis of existing academic work that maps what is already known, exposes the "knowledge gap," and concludes with a theoretical or conceptual framework guiding the study.
- **Research Methodology:**
  - *Research Design:* Specification of the overarching paradigm (e.g., Quantitative, Qualitative, Mixed-Methods) and structural approach (e.g., Experimental, Survey, Case Study).
  - *Population and Sampling:* Delineation of the target population, specific sampling techniques, and sample size justification.
  - *Data Collection Tools:* Mechanical breakdown of instruments like surveys, interview schedules, or experimental setups.
  - *Data Processing and Analysis:* An explicit plan detailing how data will be handled and evaluated (e.g., using descriptive/inferential statistics or thematic analysis).

3. The Final Matter (The Logistical Blueprint)

This section contains the supplementary documentation required to prove the study is realistic, ethical, and thoroughly credited.

- **Work Plan / Timeline:** A chronological schedule of project milestones, frequently visualized using a Gantt chart, mapping out phases from tool validation to manuscript drafting.
- **Budget and Justification:** An itemized financial breakdown tracking anticipated expenditures such as hardware, travel, software licensing, or laboratory tools.
- **References:** An exhaustive, alphabetically ordered list of every source cited, strictly formatted according to required styles like APA or IEEE.
- **Appendices:** Essential supplementary materials that would clutter the main body, including draft questionnaires, interview protocols, institutional cover letters, and informed consent forms.

Additional Expert Perspectives (Beyond the Syllabus)

While your document provides an excellent structural breakdown, robust research proposals in Computer Engineering and IT often require a few additional, highly specific components:

- **Scope and Delimitations:** Usually situated right after the objectives, this explicitly states what the research will *not* cover. In software or hardware projects, it is crucial to set boundaries so evaluators don't expect features you never intended to build.
- **Risk Management / Mitigation Plan:** IT and engineering projects are prone to technical failures, data loss, or API deprecations. A strong proposal includes a brief section predicting potential roadblocks and outlining backup plans (e.g., "If access to Database A is denied, we will utilize open-source Dataset B").
- **Expected Contributions / Deliverables:** Evaluators want to know exactly what the final output will be. Will it be a publishable paper, a working software prototype, a new algorithm, or a set of policy recommendations? Clearly defining the physical or intellectual product clarifies the project's ultimate value.

User prompt: 4.4 A case study on any research paper/project

Response: Based on the provided document, "Final BE 4 th sem(2)(1).pdf", here is the breakdown of the case study provided in Section 4.4 to illustrate how a research proposal functions in practice.

4.4 A Case Study on a Research Project

To fully understand how proposal components operate in practice, the document details a collaborative Research and Development (R&D) project model. This case study examines a multi-year agricultural R&D program focused on sustainable soil management within low-income smallholder farming systems. It follows a "Research-Development Continuum framework" to bridge physical sciences with social sciences.

1. Problem Identification & Objectives

Every successful project maps directly back to a validated real-world or theoretical issue.

- **Observed Problem:** Smallholder farmers were experiencing severe land degradation, nutrient mining, and declining crop yields caused by continuous, intensive cultivation without proper soil replenishment.
- **Project Objectives:**
  1. To determine the biophysical efficacy of integrated organic and inorganic inputs on soil health.
  2. To evaluate the socio-economic factors influencing smallholders' adoption behavior regarding these sustainable practices.

2. Methodological Implementation

To address both the biological and human elements of the problem, the project utilized a robust mixed-methods research design across three phases:

- **Phase 1 (Basic / Applied Research):** Controlled multi-locational field trials were established using a Randomized Complete Block Design (RCBD) to evaluate different treatments of organic manure and mineral fertilizers. Quantitative biophysical datasets (like soil nitrogen levels) were rigorously analyzed using Analysis of Variance (ANOVA).
- **Phase 2 (Adaptive / Action Research):** The project shifted to farmer-participatory research, piloting these agricultural technologies via on-farm experiments where local farmers actively managed treatments and provided experiential feedback.
- **Phase 3 (Socio-Economic Evaluation):** Structured cross-sectional surveys were administered, and quantitative data was processed via statistical software (like SPSS) to track adoption determinants. Simultaneously, qualitative Focus Group Discussions (FGDs) were deployed to capture human values, structural limitations, and local governance issues.

3. Outcomes & Dissemination

By mapping out how basic laboratory discoveries translate into applied field options, the project yielded measurable, real-world results:

- Targeted, data-backed fertilizer guidelines were established for regional ecosystems.
- Local agricultural capacities were built through participatory training.
- A final impact assessment provided verifiable evidence of a measurable rise in household food security.
- Findings were disseminated through peer-reviewed journals to advance scientific literature and distributed via local networks to serve the public good.

Additional Expert Perspectives (Beyond the Syllabus)

Because your course is targeted toward **BE (Computer, IT and Software)** students, the agricultural case study provided in the syllabus might feel a bit disconnected from your daily work. Here is how you can translate that exact same 3-phase mixed-methods structure into an **IT/Computer Engineering context**:

- **Translating Phase 1 (Biophysical to Computational):** Instead of testing soil nitrogen, an IT student might test *server load balancing algorithms*. You would set up a controlled, isolated network environment (your "field trial") and run quantitative benchmarking tests to measure latency, throughput, and CPU usage using tools like Apache JMeter.
- **Translating Phase 2 (Adaptive to UX/UI Testing):** Just as the agricultural researchers brought farmers in to test the fertilizer, you would deploy your software prototype to a beta-testing group of end-users. You track how they naturally interact with the UI, gathering experiential feedback and identifying bugs in real-time.
- **Translating Phase 3 (Socio-Economic to User Adoption):** Finally, you would send out surveys (using Likert scales) and conduct focus groups to understand *why* users might resist adopting your new software. You would analyze this qualitative data to uncover human-centric issues like "steep learning curves" or "privacy concerns," perfectly mirroring the mixed-methods approach of a highly professional research project.
